

JADAVPUR UNIVERSITY
B.E. ELECTRICAL ENGINEERING EXAMINATION - 2025
ELECTRICAL MACHINES - II (Part I)

Detailed Analytical Solutions provided by SR Sir

Question 1(a): "It is not absolutely necessary to laminate the rotor of an induction motor." True or False? Give reasons.

SR Sir's Thought Process

Examiners love this conceptual trap. Standard motors are laminated, but why? To reduce eddy current losses. Eddy current loss is proportional to the square of frequency (f^2). In a running induction motor, the rotor frequency is sf_1 . If slip is 3% and supply is 50Hz, rotor frequency is 1.5Hz. At this extremely low frequency, core losses are negligible. Thus, for pure running, lamination isn't strictly necessary. Solid iron rotors actually exist! However, at starting ($s = 1$), the frequency is 50Hz, so unlaminated rotors get very hot. The statement is technically True in principle, though practically laminations are used.

Answer:

True.

Reasons: The primary purpose of laminating any magnetic core is to minimize eddy current losses, which are directly proportional to the square of the operating frequency ($P_e \propto f^2$). In a 3-phase induction motor, the frequency of the magnetic flux reversals in the rotor core is the slip frequency:

$$f_2 = sf_1 \quad (1)$$

Under normal operating conditions, the slip (s) is very small (typically 2% to 5%). For a 50Hz supply, the rotor frequency f_2 is merely 1Hz to 2.5Hz. At such phenomenally low frequencies, the eddy current and hysteresis losses in the rotor core are practically negligible. Therefore, from a purely steady-state running perspective, a solid (unlaminated) iron rotor would function perfectly well without overheating.

Practical Caveat: While not "absolutely necessary," manufacturers still laminate standard rotors because, at the exact moment of starting ($s = 1$), the rotor frequency equals the full supply frequency (50Hz). A solid rotor would experience massive core losses and extreme heating during starting. However, specialized "Solid Rotor Induction Motors" do exist where extreme mechanical robustness is required, proving the statement true.

Question 1(b): What will be the direction of rotation of the rotor, if the primary winding is placed on the rotor of a poly-phase induction motor? Give reasons for your answer.

SR Sir's Thought Process

This is a recurring classic question. If primary is on the rotor and secondary (shorted) is on the stator, the RMF rotates at N_s relative to the rotor. The stator opposes the change by Lenz's Law. Since the stator can't move, the reaction pushes the rotor backwards to reduce relative speed.

Answer:

If the primary winding is placed on the rotor and the stator is short-circuited, the rotor will physically rotate in the **opposite direction** to the direction of its own rotating magnetic field (RMF).

Reasons: When the rotor (primary) is energized by a 3-phase supply, it produces a Rotating Magnetic Field (RMF) that revolves at synchronous speed (N_s) relative to the rotor structure. This moving flux cuts the stationary conductors of the short-circuited stator, inducing an EMF and consequently a current in them.

According to **Lenz's Law**, the electromagnetic torque developed acts in a direction to oppose the relative motion between the magnetic field and the stator conductors. The magnetic field attempts to drag the stator along with it to reduce this relative speed. However, the stator is physically bolted to the ground. The reaction torque therefore acts on the movable body—the rotor. To minimize the absolute speed of the magnetic field cutting the stator, the rotor is forced to physically spin backwards (opposite to its RMF), effectively reducing the net speed of the field in space.

Question 1(c): What is the relative speed between the magnetic fields produced by the stator winding and rotor winding of a 3-phase induction motor?

SR Sir's Thought Process

Another recurring kinematic question. Stator field space speed = N_s . Rotor field space speed = rotor mechanical speed + relative field speed = $N_r + (N_s - N_r) = N_s$. Both fields are stationary with respect to each other.

Answer:

The relative speed between the magnetic fields produced by the stator winding and the rotor winding is **Zero**.

Explanation:

1. The balanced 3-phase stator winding produces a stator magnetic field that rotates at synchronous speed (N_s) with respect to the stationary stator frame (in space).
2. The rotor rotates at a mechanical speed N_r . The relative speed of the stator field cutting the rotor is sN_s . This induces rotor currents at slip frequency (sf_1).
3. These slip-frequency currents generate a rotor magnetic field that rotates at speed sN_s *with respect to the rotor body*.
4. The absolute speed of the rotor magnetic field in space is the vector sum of the rotor's mechanical speed and the field's relative speed:

$$\text{Space Speed of Rotor Field} = N_r + sN_s = (1 - s)N_s + sN_s = N_s \quad (2)$$

Because both magnetic fields rotate at the exact same speed (N_s) and in the exact same direction in space, they remain magnetically locked. Their relative speed to one another is zero.

Question 1(d): Find an expression for distribution factor of a distributed winding.

SR Sir's Thought Process

Need to define distribution factor K_d as phasor sum over arithmetic sum. Assume m slots per pole per phase, slot angle γ . Draw an arc of phasors forming a polygon. The arithmetic sum is mE_c . The phasor sum is the chord subtending angle $m\gamma$.

Answer:

Distribution Factor (K_d): In a distributed winding, the coils of a given phase are spread across multiple adjacent slots rather than being concentrated in a single slot. Because of this spatial displacement, the EMFs induced in individual coils are not perfectly in phase. The Distribution Factor (K_d) is defined as:

$$K_d = \frac{\text{Phasor sum of coil EMFs}}{\text{Arithmetic sum of coil EMFs}} \quad (3)$$

Derivation: Let: m = number of slots per pole per phase γ = electrical angle between adjacent slots (slot pitch) E_c = EMF induced in each individual coil side

Geometrically, the individual coil EMFs can be represented as consecutive chords of a circle with radius R and center O . Each chord (coil EMF) subtends an angle γ at the center. The magnitude of a single coil EMF is:

$$E_c = 2R \sin\left(\frac{\gamma}{2}\right) \quad (4)$$

The arithmetic sum of the EMFs in m coils is:

$$E_{arithmetic} = mE_c = m \left[2R \sin\left(\frac{\gamma}{2}\right) \right] \quad (5)$$

The resultant EMF is the phasor sum of these m coils. Geometrically, this is the single large chord connecting the start of the first phasor to the end of the last phasor. This grand chord subtends a total angle of $m\gamma$ at the center O . The magnitude of the resultant phasor sum is:

$$E_{phasor} = 2R \sin\left(\frac{m\gamma}{2}\right) \quad (6)$$

Taking the ratio gives the distribution factor:

$$K_d = \frac{E_{phasor}}{E_{arithmetic}} = \frac{2R \sin\left(\frac{m\gamma}{2}\right)}{m \left[2R \sin\left(\frac{\gamma}{2}\right) \right]} \quad (7)$$

$$\boxed{K_d = \frac{\sin\left(\frac{m\gamma}{2}\right)}{m \sin\left(\frac{\gamma}{2}\right)}} \quad (8)$$

Question 1(e): Describe the phenomenon of cogging and crawling.

SR Sir's Thought Process

Cogging: Magnetic locking at start when Stator slots $S_1 =$ Rotor slots S_2 or integral multiples. Reluctance is minimum, motor refuses to spin. Crawling: Space harmonics (specifically 7th) create a forward torque dip at $N_s/7$. The motor gets "stuck" at this low speed instead of accelerating to rated speed.

Answer:

1. Cogging (Magnetic Locking): Cogging is a phenomenon where a squirrel cage induction motor refuses to start at all, remaining stationary and humming loudly when the supply is switched on.

- **Cause:** It occurs when the number of stator slots (S_1) is equal to, or an integral multiple of, the number of rotor slots (S_2).
- **Mechanism:** When the slots align, the air gap reluctance between the stator teeth and rotor teeth reaches a minimum. The magnetic system prefers this state of minimum reluctance, creating a strong alignment force (reluctance torque) that opposes rotation. If this reluctance torque exceeds the starting electromagnetic torque, the rotor magnetically "locks" to the stator.
- **Prevention:** It is eliminated by skewing the rotor slots and ensuring $S_1 \neq S_2$.

2. Crawling: Crawling is a phenomenon where the induction motor starts but fails to accelerate to its full synchronous speed, instead running stably at a very low speed (typically one-seventh of synchronous speed).

- **Cause:** It is caused by the presence of space harmonics in the air gap flux wave, primarily the 7th harmonic.
- **Mechanism:** The 7th space harmonic acts like a separate 3-phase machine with 7 times the poles, creating a forward-rotating magnetic field that spins at $N_s/7$. This harmonic field produces its own parasitic torque-speed curve that peaks near $N_s/7$ and then drops sharply, creating a "dip" or "saddle" in the total resultant motor torque curve. If the mechanical load torque intersects this dip, the motor cannot accelerate past it and will "crawl" at $\approx N_s/7$.

Question 1(f): What is deep bar rotor induction motor? How it produces high starting torque?

SR Sir's Thought Process

Re-iterating the Skin Effect answer from previous years. Deep, narrow slots. High leakage flux at bottom. At $s = 1$, frequency is 50Hz, current forced to top, high R_{eff} . At running, frequency is low, current uses full bar, low R_{eff} .

Answer:

Deep Bar Rotor Induction Motor: A deep bar rotor is a specialized construction of a squirrel cage induction motor where the rotor slots are punched deep into the iron core, and the conductive bars placed inside have a high depth-to-width ratio. It is designed to automatically provide high starting torque while maintaining high running efficiency, without needing external slip rings.

Mechanism of High Starting Torque (Skin Effect): The operation relies on the **Skin Effect**, driven by the changing frequency of rotor currents. The bottom layer of the deep bar is deeply embedded in the iron core and links with much more slot leakage flux than the top layer near the air gap. Consequently, the leakage inductance (L) and reactance ($X = 2\pi fL$) of the bottom is much higher than the top.

- **At Starting ($s = 1$):** The rotor frequency is 50Hz. At this high frequency, the reactance of the bottom layer is so immense that the AC current is choked off and forced entirely into the narrow top layer of the bar. This severe current crowding heavily restricts the effective cross-sectional area, making the effective rotor resistance (R_2) very high. Since starting torque $T_{st} \propto R_2$, a massive starting torque is developed.
- **At Running ($s \approx 0.03$):** The rotor frequency is very low ($\approx 1.5\text{Hz}$). The reactance term vanishes. Current distributes evenly across the entire deep bar. The full cross-sectional area is utilized, yielding a low DC resistance, which minimizes copper losses and maximizes efficiency.

Question 2(a): Explain how a poly-phase winding fed from a balanced poly-phase supply produces a rotating magnetic field with constant amplitude.

SR Sir's Thought Process

Standard trigonometric proof for RMF. 3-phase currents, 3 spatial axes at 120° . Forward wave adds up, backward wave cancels to zero. Result is $1.5F_m$.

Answer:

A Rotating Magnetic Field (RMF) of constant amplitude is produced due to the precise spatial and temporal displacement of the machine's windings and supply currents.

Mathematical Explanation: Consider a stator with three identical windings (Phases A, B, C) physically displaced from each other by 120° electrical in space. When connected to a balanced 3-phase AC supply, the instantaneous currents are displaced in time by 120° :

$$i_A(t) = I_m \cos(\omega t) \quad (9)$$

$$i_B(t) = I_m \cos(\omega t - 120^\circ) \quad (10)$$

$$i_C(t) = I_m \cos(\omega t + 120^\circ) \quad (11)$$

Each phase current generates a pulsating Magnetomotive Force (MMF) along its respective spatial axis. Let θ be the spatial angle measured from the axis of Phase A. The spatial MMF waves are:

$$F_A(\theta, t) = F_m \cos(\omega t) \cos(\theta) \quad (12)$$

$$F_B(\theta, t) = F_m \cos(\omega t - 120^\circ) \cos(\theta - 120^\circ) \quad (13)$$

$$F_C(\theta, t) = F_m \cos(\omega t + 120^\circ) \cos(\theta + 120^\circ) \quad (14)$$

where F_m is the peak MMF of a single phase.

The total resultant MMF in the air gap (F_{res}) is the sum of these three fields. Using the trigonometric identity $\cos X \cos Y = \frac{1}{2}[\cos(X - Y) + \cos(X + Y)]$:

$$F_A = \frac{F_m}{2} [\cos(\omega t - \theta) + \cos(\omega t + \theta)] \quad (15)$$

$$F_B = \frac{F_m}{2} [\cos(\omega t - \theta) + \cos(\omega t + \theta - 240^\circ)] \quad (16)$$

$$F_C = \frac{F_m}{2} [\cos(\omega t - \theta) + \cos(\omega t + \theta + 240^\circ)] \quad (17)$$

Summing these up ($F_{res} = F_A + F_B + F_C$):

$$F_{res}(\theta, t) = \frac{3}{2} F_m \cos(\omega t - \theta) + \frac{F_m}{2} [\cos(\omega t + \theta) + \cos(\omega t + \theta - 240^\circ) + \cos(\omega t + \theta + 240^\circ)] \quad (18)$$

The terms in the second bracket represent three balanced co-phasal vectors separated by exactly 240° (which acts identically to 120°). The sum of any three balanced sinusoidal quantities is zero. Thus, the backward-traveling wave vanishes completely.

$$\boxed{F_{res}(\theta, t) = 1.5F_m \cos(\omega t - \theta)} \quad (19)$$

This resultant equation represents a traveling magnetic wave with a strictly **constant amplitude** of $1.5F_m$ that rotates through space at a constant synchronous angular velocity ω .

Question 2(b): Derive an expression for power developed in an induction motor. Also derive the condition for maximum power and draw the power-speed characteristics.

SR Sir's Thought Process

Note that this asks for POWER, not torque. 1. Equation for gross mechanical power $P_d = 3I_2'^2 R_2' \frac{1-s}{s}$. 2. Max power condition: apply maximum power transfer theorem to the equivalent circuit where $R_L = R_2' \frac{1-s}{s}$ is the variable load. Maximum power is transferred when $R_L = |Z_{th}| = \sqrt{R_1^2 + (X_1 + X_2')^2}$. 3. Equate $R_2' \frac{1-s_{mP}}{s_{mP}} = |Z_{eq}|$ and solve for s_{mP} . 4. Draw Power vs Speed curve. Power is 0 at $N_r = 0$ and $N_r = N_s$, peaking in between.

Answer:

1. Expression for Power Developed: From the exact per-phase equivalent circuit of an induction motor referred to the stator, the gross mechanical power developed (P_d) is the power consumed by the fictitious mechanical load resistance $R_L = R_2' \left(\frac{1-s}{s}\right)$. For all 3 phases:

$$P_d = 3(I_2')^2 R_2' \left(\frac{1-s}{s}\right) \quad (20)$$

The magnitude of the rotor current I_2' is:

$$I_2' = \frac{V_1}{\sqrt{\left(R_1 + \frac{R_2'}{s}\right)^2 + (X_1 + X_2')^2}} \quad (21)$$

Substituting I_2' yields the exact expression for developed mechanical power:

$$P_d = \frac{3V_1^2 R_2' \left(\frac{1-s}{s}\right)}{\left(R_1 + \frac{R_2'}{s}\right)^2 + (X_1 + X_2')^2} \text{ Watts} \quad (22)$$

2. Condition for Maximum Power: To find the slip at which maximum mechanical power is developed (s_{mP}), we apply the **Maximum Power Transfer Theorem** to the equivalent circuit. The theorem states that maximum power is transferred to a variable load resistance when it equals the magnitude of the Thevenin equivalent impedance looking back into the source. Here, the variable load is $R_L = R_2' \left(\frac{1-s}{s}\right)$, and the fixed impedance is $Z_{th} = R_1 + j(X_1 + X_2')$. Thus, for maximum power:

$$R_2' \left(\frac{1-s_{mP}}{s_{mP}}\right) = \sqrt{R_1^2 + (X_1 + X_2')^2} \quad (23)$$

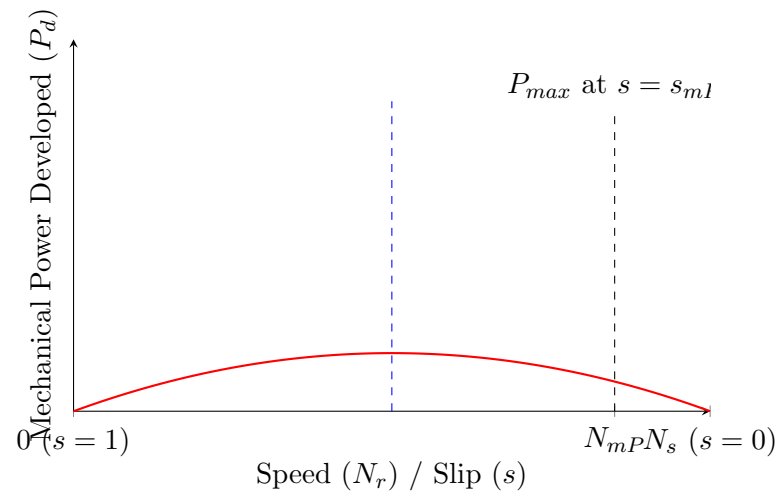
Let the total leakage impedance magnitude be $|Z_{eq}| = \sqrt{R_1^2 + (X_1 + X_2')^2}$.

$$\frac{1}{s_{mP}} - 1 = \frac{|Z_{eq}|}{R_2'} \implies \frac{1}{s_{mP}} = \frac{R_2' + |Z_{eq}|}{R_2'} \quad (24)$$

$$s_{mP} = \frac{R_2'}{R_2' + \sqrt{R_1^2 + (X_1 + X_2')^2}} \quad (25)$$

Note that the slip for maximum power (s_{mP}) is different from the slip for maximum torque (s_{mT}). Maximum power occurs at a higher speed (lower slip) than maximum torque.

3. Power-Speed Characteristics: At standstill ($N_r = 0, s = 1$), $P_d = 0$ (no mechanical work is done). At synchronous speed ($N_r = N_s, s = 0$), $P_d = 0$ (no induced current). The power peaks at speed $N_r = (1 - s_{mP})N_s$.



Question 3(a): Develop the equivalent circuit of an induction motor from the basic principle.

SR Sir's Thought Process

1. Start with transformer logic.
2. Write relative slip frequency logic (sE_2, sX_2).
3. Write rotor current equation.
4. Divide by s to make a stationary equivalent load.
5. Refer to stator. Draw circuit.

Answer:

An induction motor fundamentally operates as a rotating transformer where the stator is the primary and the short-circuited rotor is the secondary.

Step 1: The Rotating Rotor Circuit Let V_1 be the stator phase voltage, R_1 the stator resistance, and X_1 the stator leakage reactance. The no-load core flux and losses are modeled by a parallel R_c and X_m branch. At standstill ($s = 1$), let the induced EMF in the rotor be E_2 and its leakage reactance be X_2 . When the rotor spins at slip s , the relative speed of the magnetic field cutting the rotor conductors drops. Consequently:

- Induced rotor voltage becomes: $E_{2s} = sE_2$
- Rotor leakage reactance becomes: $X_{2s} = sX_2$ (since frequency $f_2 = sf_1$)

The actual rotor current (I_2) flowing through the physical rotor resistance R_2 is:

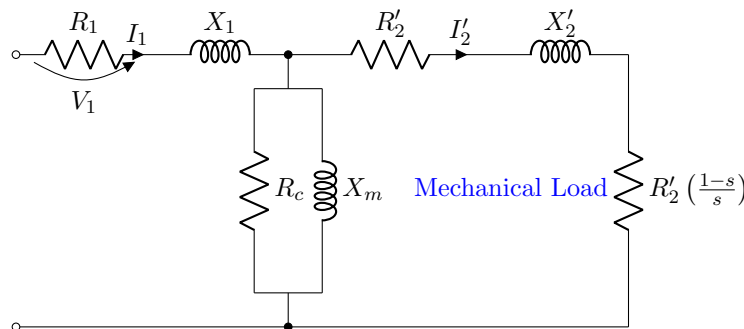
$$I_2 = \frac{sE_2}{\sqrt{R_2^2 + (sX_2)^2}} \quad (26)$$

Step 2: Mathematical Transformation to a Stationary Circuit Equation (26) represents a rotating circuit with variable frequency. To couple this directly with the stationary 50Hz stator circuit, we divide both the numerator and denominator by s :

$$I_2 = \frac{E_2}{\sqrt{\left(\frac{R_2}{s}\right)^2 + X_2^2}} \quad (27)$$

This mathematically converts the rotating secondary into a purely stationary secondary where the voltage is fixed at E_2 , the reactance is fixed at X_2 , and the resistance becomes a variable parameter $\frac{R_2}{s}$.

Step 3: Referring to the Stator We refer these stationary rotor parameters back to the primary stator side using the effective turns ratio (k). $R'_2 = k^2 R_2$, $X'_2 = k^2 X_2$. The equivalent variable resistance $\left(\frac{R'_2}{s}\right)$ is then split into two distinct physical components: $\frac{R'_2}{s} = R'_2 + R'_2 \left(\frac{1-s}{s}\right)$. The first term represents actual copper loss. The second term represents the mechanical power output. Integrating this with the stator gives the exact equivalent circuit:



Question 3(b): How the equivalent circuit parameters of an induction motor are determined in the laboratory?

SR Sir's Thought Process

Standard tests: DC resistance test, No-Load test, Blocked Rotor test. Briefly describe procedure and extracted formulas.

Answer:

The exact equivalent circuit parameters are determined experimentally using three standard laboratory tests:

1. DC Resistance Test (for Stator Resistance R_1) A low DC voltage is applied across two stator terminals to circulate rated current. The resistance is calculated using Ohm's Law ($R_{dc} = V_{dc}/2I_{dc}$ for a star connection). Because AC current experiences the skin effect, this DC resistance is scaled up by a factor of 1.2 to 1.5 to obtain the effective AC stator resistance (R_1).

2. No-Load Test (for Shunt Parameters R_c, X_m) The motor is operated at rated voltage (V_0) and rated frequency with no mechanical load on the shaft. It spins very close to synchronous speed ($s \approx 0$). Consequently, the mechanical load branch $R'_2(\frac{1-s}{s})$ approaches infinity, acting as an open circuit.

- Measured: Line voltage V_0 , Line current I_0 , 3-Phase Power P_0 .
- Extracted: No-load power factor $\cos \phi_0 = P_0/(\sqrt{3}V_0I_0)$.
- Core loss resistor: $R_c = V_{0ph}/(I_0 \cos \phi_0)$.
- Magnetizing reactance: $X_m = V_{0ph}/(I_0 \sin \phi_0)$.

3. Blocked Rotor Test (for Series Parameters R'_2, X_{eq}) The rotor is physically locked ($s = 1$). A vastly reduced AC voltage (V_{br}) is applied until rated full-load current (I_{br}) flows. At such low voltage, core losses and magnetizing current are negligible, so the shunt branch is ignored.

- Measured: Line voltage V_{br} , Line current I_{br} , 3-Phase Power P_{br} .
- Equivalent resistance: $R_{br} = (P_{br}/3)/I_{br,ph}^2$.
- Equivalent impedance: $Z_{br} = V_{br,ph}/I_{br,ph}$.
- Equivalent reactance: $X_{br} = \sqrt{Z_{br}^2 - R_{br}^2}$.
- Extracted: Rotor resistance $R'_2 = R_{br} - R_1$. Leakage reactances are assumed evenly split: $X_1 = X'_2 = X_{br}/2$.

Question 4(a): Describe different starting methods of a 3-phase induction motor.

SR Sir's Thought Process

List methods for SQIM (DOL, Stator resistance, Auto-transformer, Star-Delta) and SRIM (Rotor resistance). Briefly describe the mechanism of each.

Answer:

Starting a 3-phase induction motor directly from the grid results in massive starting currents (5-7 times full load current) because there is no back-EMF at standstill. Starters are used to limit this inrush current.

Methods for Squirrel Cage Induction Motors (SQIM):

1. **Direct-On-Line (DOL) Starting:** The motor is connected directly to full supply voltage. Only used for very small motors (usually < 5 HP) where the grid can absorb the starting current surge without significant voltage dips.
2. **Stator Resistance / Reactor Starting:** External resistors or inductors are inserted in series with the stator windings to drop a portion of the supply voltage during start-up. Once the motor accelerates, these are bypassed. It is cheap but causes high I^2R power losses in the resistors.
3. **Auto-Transformer Starting:** An auto-transformer with tapplings (e.g., 60%, 80%) is used to apply a reduced voltage to the motor. This method is highly efficient as it reduces the line current drawn from the grid by the square of the tapping ratio. It provides the best torque per ampere of line current but is expensive.
4. **Star-Delta ($Y - \Delta$) Starting:** The most common method. A switch initially connects the stator windings in Star, reducing the phase voltage to $1/\sqrt{3}$ of normal. Consequently, starting line current drops to $1/3$ of DOL current. Near full speed, the switch reconnects the windings in Delta for normal operation.

Method for Slip Ring Induction Motors (SRIM):

5. **Rotor Resistance Starting:** External variable resistors are connected to the rotor circuit via slip rings. This limits the starting current while simultaneously vastly increasing the starting torque ($T_{st} \propto R_2$). As the motor speeds up, the external resistance is gradually cut out until the rings are short-circuited.

Question 4(b): What are the advantages and disadvantages of a SRIM over a SQIM?

SR Sir's Thought Process

SRIM = Slip Ring (Wound Rotor). SQIM = Squirrel Cage. Advantages: High starting torque, low starting current, speed control via rotor resistance. Disadvantages: Higher cost, maintenance of brushes, slightly lower efficiency (friction and copper losses), lower power factor.

Answer:

A Slip Ring Induction Motor (SRIM) has a wound rotor connected to external slip rings, whereas a Squirrel Cage Induction Motor (SQIM) has permanently shorted solid metal bars.

Advantages of SRIM over SQIM:

- **Excellent Starting Characteristics:** By inserting external resistance into the rotor circuit during start-up, a SRIM can develop massive starting torque (up to maximum pull-out torque) while simultaneously drawing a very low starting current from the grid.
- **Speed Control Capability:** The speed of a SRIM can be controlled below synchronous speed by varying the external rotor resistance, which modifies the torque-slip characteristics dynamically.
- **High Inertia Starting:** SRIMs are highly capable of smoothly accelerating massive, high-inertia loads (like crushers, hoists, and heavy mills) without overheating the internal windings, as heat is dissipated in the external resistors.

Disadvantages of SRIM over SQIM:

- **Higher Initial Cost & Complexity:** The wound rotor, slip rings, and brush assemblies make SRIMs significantly more expensive and complex to manufacture than the rugged, simple SQIM.
- **High Maintenance:** The brushes and slip rings are prone to wear, sparking, and require regular inspection and replacement. SQIMs are practically maintenance-free.
- **Lower Efficiency:** SRIMs typically have higher rotor copper losses and additional mechanical friction losses due to brush contact, resulting in a slightly lower overall efficiency compared to an equivalent SQIM.
- **Lower Power Factor:** The extra winding complexity slightly increases leakage reactance, leading to a poorer operating power factor.

Question 5(a): A 50 kW, 400 volt, 3-phase, 4-pole, 50 Hz wound rotor induction motor develops a maximum internal torque of 240% at a slip of 20% when working under rated voltage and frequency conditions and with slip ring short-circuited. Determine (a) full load slip, (b) rotor copper loss at full load and (c) starting torque in N-m. Neglect stator resistance and rotational losses.

SR Sir's Thought Process

Given: $P_{out} = 50 \text{ kW}$. $T_{max}/T_{fl} = 2.4$. $s_{mT} = 0.2$. $R_1 = 0$, $P_{rot} = 0 \Rightarrow P_{dev} = 50 \text{ kW}$. Use torque ratio equation: $T/T_{max} = 2/(s/s_{mT} + s_{mT}/s)$. Solve for s_{fl} . Rotor Cu loss $P_{rc} = P_{dev} \times (s/(1-s))$. Starting torque $T_{st} = T_{max} \times 2/(1/s_{mT} + s_{mT}/1)$. Need T_{max} in N-m first. Find $T_{fl} = P_{dev}/\omega_r$. $T_{max} = 2.4T_{fl}$.

Answer:

Given Data: Output Power $P_{out} = 50 \text{ kW} = 50000 \text{ W}$. Max Torque $T_{max} = 2.4 \times T_{fl}$ (where T_{fl} is full load torque). Slip at max torque $s_{mT} = 20\% = 0.2$. Poles $P = 4$, Frequency $f = 50 \text{ Hz}$. Since rotational losses are neglected, gross mechanical power developed $P_{dev} = P_{out} = 50000 \text{ W}$.

(a) Full Load Slip (s_{fl}): Since stator resistance is neglected, the torque ratio formula is:

$$\frac{T_{fl}}{T_{max}} = \frac{2}{\frac{s_{fl}}{s_{mT}} + \frac{s_{mT}}{s_{fl}}} \quad (28)$$

Substitute the known values:

$$\frac{1}{2.4} = \frac{2}{\frac{s_{fl}}{0.2} + \frac{0.2}{s_{fl}}} = \frac{2}{5s_{fl} + \frac{0.2}{s_{fl}}} \quad (29)$$

Cross-multiply:

$$5s_{fl} + \frac{0.2}{s_{fl}} = 4.8 \quad (30)$$

Multiply by s_{fl} to form a quadratic equation:

$$5s_{fl}^2 - 4.8s_{fl} + 0.2 = 0 \quad (31)$$

Using the quadratic formula:

$$s_{fl} = \frac{4.8 \pm \sqrt{(4.8)^2 - 4(5)(0.2)}}{2(5)} = \frac{4.8 \pm \sqrt{23.04 - 4}}{10} = \frac{4.8 \pm \sqrt{19.04}}{10} \quad (32)$$

$$s_{fl} = \frac{4.8 \pm 4.3635}{10} \quad (33)$$

We take the smaller root for normal operating region:

$$s_{fl} = \frac{4.8 - 4.3635}{10} = 0.04365 \quad (34)$$

Full load slip is 0.04365 (or 4.365%).

(b) Rotor Copper Loss at Full Load (P_{rc}): The relationship between developed mechanical power and rotor copper loss is:

$$P_{rc} = P_{dev} \left(\frac{s_{fl}}{1 - s_{fl}} \right) \quad (35)$$

$$P_{rc} = 50000 \times \left(\frac{0.04365}{1 - 0.04365} \right) = 50000 \times \left(\frac{0.04365}{0.95635} \right) = 50000 \times 0.04564 \quad (36)$$

$$P_{rc} = \mathbf{2282.1 \text{ Watts}} \quad (37)$$

(c) Starting Torque in N-m (T_{st}): First, find synchronous speed ω_s :

$$N_s = \frac{120 \times 50}{4} = 1500 \text{ RPM} \implies \omega_s = \frac{2\pi \times 1500}{60} = 157.08 \text{ rad/s} \quad (38)$$

Full load rotor speed ω_r :

$$\omega_r = \omega_s(1 - s_{fl}) = 157.08 \times (1 - 0.04365) = 150.22 \text{ rad/s} \quad (39)$$

Full load torque T_{fl} :

$$T_{fl} = \frac{P_{dev}}{\omega_r} = \frac{50000}{150.22} = 332.84 \text{ N-m} \quad (40)$$

Maximum torque T_{max} :

$$T_{max} = 2.4 \times T_{fl} = 2.4 \times 332.84 = 798.82 \text{ N-m} \quad (41)$$

Now, find starting torque ($s = 1$) using the ratio formula:

$$\frac{T_{st}}{T_{max}} = \frac{2}{\frac{1}{s_{mT}} + \frac{s_{mT}}{1}} = \frac{2}{\frac{1}{0.2} + 0.2} = \frac{2}{5 + 0.2} = \frac{2}{5.2} = 0.3846 \quad (42)$$

$$T_{st} = 0.3846 \times 798.82 = \mathbf{307.23 \text{ N-m}} \quad (43)$$

Question 5(b): A 15 kW, 3-phase, 4 pole, 50 Hz, induction motor has a full load slip of 0.03. Mechanical and stray load losses at full load are 3.5% of output power. Compute i) Power delivered by stator to rotor ii) Electromagnetic torque at full load iii) Rotor ohmic loss at full load.

SR Sir's Thought Process

$P_{out} = 15000$ W. Losses $= 0.035 \times 15000 = 525$ W. $P_{dev} = P_{out} + P_{losses} = 15525$ W. P_{ag} (stator to rotor) $= P_{dev}/(1 - s) = 15525/0.97 = 16005.15$ W. $T_{em} = P_{ag}/\omega_s$. $\omega_s = 2\pi \times 1500/60 = 157.08$. $T_{em} = 101.89$ N-m. $P_{rc} = s \times P_{ag} = 0.03 \times 16005.15 = 480.15$ W.

Answer:

Given Data: Output Power $P_{out} = 15$ kW $= 15000$ W Full load slip $s = 0.03$ Mechanical and stray losses $P_{mech} = 3.5\%$ of $P_{out} = 0.035 \times 15000 = 525$ W.

Preliminary Calculation: Gross Mechanical Power Developed (P_d) The total mechanical power generated by the rotor must supply both the useful output power and the internal mechanical friction/stray losses.

$$P_d = P_{out} + P_{mech} = 15000 + 525 = 15525 \text{ W} \quad (44)$$

i) Power delivered by stator to rotor (Air-Gap Power, P_{ag}): The mechanical power developed is related to the power transferred across the air gap by the fundamental relation $P_d = (1 - s)P_{ag}$.

$$P_{ag} = \frac{P_d}{1 - s} = \frac{15525}{1 - 0.03} = \frac{15525}{0.97} = \mathbf{16005.15 \text{ W}} \quad (45)$$

ii) Electromagnetic Torque at full load (T_{em}): Synchronous speed in RPM (N_s):

$$N_s = \frac{120f}{P} = \frac{120 \times 50}{4} = 1500 \text{ RPM} \quad (46)$$

Synchronous angular velocity (ω_s):

$$\omega_s = \frac{2\pi N_s}{60} = \frac{2\pi \times 1500}{60} = 157.08 \text{ rad/sec} \quad (47)$$

The electromagnetic torque is the air-gap power divided by the synchronous speed:

$$T_{em} = \frac{P_{ag}}{\omega_s} = \frac{16005.15}{157.08} = \mathbf{101.89 \text{ N-m}} \quad (48)$$

iii) Rotor Ohmic Loss at full load (P_{rc}): The rotor copper (ohmic) loss is the fraction of air-gap power dissipated as heat, determined directly by the slip $P_{rc} = sP_{ag}$.

$$P_{rc} = 0.03 \times 16005.15 = \mathbf{480.15 \text{ W}} \quad (49)$$

(Alternatively: $P_{rc} = P_{ag} - P_d = 16005.15 - 15525 = 480.15 \text{ W}$)

— END OF 2025 PART I SOLUTIONS —
May you master these concepts with absolute clarity.